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Hierarchical Reinforcement Learning for Long-Distance Multi-Objective Optimization of Heavy-Haul Trains via Sequential Task Assignment Across Spatial Segments

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ABSTRACT

The optimization of long-distance multi-objective operations for heavy-haul trains faces low solution efficiency when applying intelligent algorithms. Currently, few studies consider the changing emphasis of multiple operational objectives, such as safety, stability, punctuality and energy efficiency changes across different operational segments through task switching. To address this issue, this paper proposes a sequential multi-task collaboration hierarchical reinforcement learning framework for dynamic multi-objective optimization in long-distance operations. First, an event-driven time-dependent classification model is established to decompose the long-distance optimization problem into multiple subtasks, thereby reducing optimization complexity. Furthermore, a hierarchical reinforcement learning method based on sequential multi-task collaboration is designed, in which segment-specific subtask policies are trained in parallel and invoked sequentially along the route, while global coordination is achieved through a top-level dynamic programming framework. Simulation results based on a real 340-km railway line demonstrate that, compared to a single-agent global optimization method, the proposed framework improves convergence speed by nearly 40% and achieves superior optimization quality. It effectively restricts coupler forces within the safe limit of 1100 kN, while successfully balancing multiple objectives including speed tracking and energy consumption.

1 | Introduction

Heavy-haul trains (HHTs) play a critical role in modern freight transportation systems, particularly in the mining and bulk cargo sectors. With train configurations often exceeding 20,000 tons in weight and extending up to 2 km in length, their unique operational characteristics coupled with stringent safety and stability requirements pose significant control challenges when operating on long steep downgrades.

Specifically, as large-inertia systems, HHTs accumulate substantial kinetic energy during descent, which cannot be sufficiently managed by electric braking alone, necessitating coordinated electro-pneumatic braking for effective speed control. Without a well-coordinated braking strategy, there is a heightened risk of safety incidents such as overspeeding. Moreover, the extended train length leads to complex longitudinal dynamic behaviours, where uneven distribution of coupler forces may cause in-train oscillations or even coupler failures, directly jeopardizing operational stability and structural integrity. Additionally, given the long-distance nature of heavy-haul routes, environmental conditions vary over time, requiring adaptive control strategies. Balancing multiple objectives, including safety, stability, punctuality and energy efficiency, has thus become a central challenge in enhancing the operational performance of HHTs. In recent studies, similar challenges in optimizing complex systems have been tackled using data-driven models [1].

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Addressing the operational challenges of HHTs mentioned above, it is of great significance to conduct multi-objective operation optimization to reduce longitudinal in-train forces [2] and enhance multi-objective operational performance. Similar approaches have been applied in energy systems, where predictive control strategies are used to optimize load coordination, which has proven effective in improving system efficiency and stability [3]. Early research on train operation optimization primarily involved constructing a Hamiltonian function through equations of motion and objective functions, and solving the optimal control problem using variational methods based on Pontryagin's maximum principle. In 1996, Howlett from the University of South Australia achieved the solution of this optimization problem under simple constraints [4], demonstrating the fundamental optimal regime sequence (full-power traction, constant speed, coasting and maximum braking), and further proposed the local energy minimization principle [5], proving the uniqueness of optimal switching [6, 7]. Recent scholars have further considered optimal connections under complex constraints, such as train traction system efficiency and pantograph power [8].

This method requires rigorous mathematical proofs. As the considered factors become increasingly complex, it becomes difficult to apply to intricate optimization problems. Consequently, researchers have turned to mathematical programming-based algorithms to solve the speed profile optimization problem, transforming constraints and objectives into standard forms for mathematical programming algorithms (e.g., quadratic, linear forms) for solution.

Without applying air braking, optimization methods primarily target gentle gradients, including model predictive control [9] and fuzzy control [10]. These methods iteratively calculate longitudinal forces during operation to solve for the optimal speed profile, thereby reducing longitudinal impacts and improving average speed. However, suppressing longitudinal forces under cyclical air braking conditions on long, steep downhill sections is more challenging. Scholars have addressed this by establishing air brake models with simple functional forms and using programming algorithms to solve cooperative strategies for air braking and traction/electric braking control inputs [11]. Common methods include quadratic programming [12], mixed-integer programming [13, 14] and fuzzy control [15].

Although programming methods can achieve efficient solutions by formulating optimization models, considering complex objectives and models often necessitates model order reduction for high-order constraints and models, making it difficult to guarantee solution accuracy. This is particularly true for air brake systems, whose low-order models struggle to accurately describe time delays, propagation, and performance degradation characteristics. In light of this, some researchers have investigated the effects of various factors on air brake performance, such as brake shoe temperature [16], train formation [17] and brake cylinder filling rate [18]. Wu provided a review of different air brake models [19]. The aforementioned models are too complex for direct use within programming methods. In recent years, intelligent algorithms have become popular approaches for handling such optimization problems, as they do not require order reduction techniques, like linearization or complex mathematical derivations, and can directly incorporate complex models and constraints. These include, but are not limited to, various genetic algorithms [20–22], neural networks [23, 24], machine learning [25, 26], dynamic programming [27] and reinforcement learning (RL) [28–31]. Among them, reinforcement learning is widely applied in operation optimization as it can learn sequential decisions through continuous trial-and-error interaction in complex or even model-free environments.

In recent years, reinforcement learning has achieved significant progress in the broader field of complex traffic and vehicle control. In particular, research on urban traffic signal control has gradually evolved from isolated single-intersection decision making toward regional, hierarchical and heterogeneous cooperative control, thereby improving global traffic efficiency while enhancing local adaptability [32, 33]. Meanwhile, the optimization objectives of traffic control have expanded beyond conventional efficiency-oriented formulations to unified multi-objective reward modelling that explicitly incorporates low-carbon performance, safety and operational robustness. Recent studies have shown that distributed reinforcement learning can be used for cooperative low-carbon traffic network optimization, while signal control can also be coordinated with connected and automated vehicles to jointly improve safety and efficiency at intersections [34, 35]. In vehicle control, hierarchical reinforcement learning and game-theoretic learning architectures have been increasingly adopted for autonomous driving tasks that jointly optimize energy consumption and traffic efficiency [36]. Overall, the research frontier is progressively shifting toward hierarchical reinforcement learning, distributed learning, multi-objective reward shaping under safety and efficiency constraints and communication-efficient cooperative optimization mechanisms [37, 38], which provide important methodological support for the sequential multi-task collaboration hierarchical reinforcement learning (HRL) framework proposed in this paper.

It is important to note that HHTs typically operate over large spatial distances with complex environments and train models. Using intelligent algorithms to solve such optimization problems often involves large solution scales and high dimensionality. Taking reinforcement learning as an example, when solving long-distance, large-scale optimization problems, its exploration mechanism often struggles to effectively capture long-term return maximization, easily falls into local optima and its performance can vary significantly across different solution attempts.

Sequential multi-task collaboration provides a practical paradigm for solving such large-scale problems. It emphasizes coordinated learning and execution across multiple subtasks within a unified hierarchical framework [39]. By decomposing the global optimization problem into several coordinated subtasks and assigning each subtask a dedicated learned policy, this paradigm can significantly improve solution efficiency for large-scale problems and alleviate the curse of dimensionality [40, 41]. Related task-decomposition and collaborative scheduling ideas have been widely applied in railway multi-train cooperative scheduling [42, 43] and urban traffic scheduling [44, 45]. The concept of self-organization in complex systems, such as the emergence of lane formations in bidirectional pedestrian flow, further suggests that local task-level coordination can induce globally desirable behaviour, which provides useful inspiration for railway operation optimization [46]. Although such

sequential collaborative task execution has not yet been applied to single-train long-distance operation optimization, these important innovations provide valuable research insights and solid empirical foundations.

Based on the above analysis, this paper proposes a sequential multi-task collaboration HRL (MTC-HRL) framework suitable for long-distance operation optimization of HHTs. For clarity, the proposed framework organizes the route into segment-specific subtasks. The corresponding subtask policies are trained in parallel offline and invoked sequentially during online decision-making according to the train's spatial progression, so the method is accurately characterized as a sequential multi-task collaboration mechanism. The aim is to generate optimal speed profiles and a combined air and electric braking strategy, balancing solution accuracy and computational efficiency. Addressing the differences in operation modes and optimization objectives across various operational scenarios over long distances, the problem is modelled as a set of segment-specific subtasks. Each subtask is associated with a dedicated policy for offline learning, and a top-level decision-making algorithm is designed to achieve optimal subtask allocation over the long distance, generating a globally optimized policy. The main contributions of this paper are as follows:

1. A sequential MTC-HRL method suitable for long-distance operation optimization of HHT is proposed. Compared to single-agent deep reinforcement learning and other intelligent algorithms, it reduces problem complexity and improves solution efficiency while maintaining accuracy.
2. A task classification model has been designed to distinguish the operational modes between long steep downhill sections and undulating gradient sections. Different tasks are assigned tailored optimization objectives and differentiated weights, including speed tracking, energy efficiency and coupler force stability. Compared to traditional hierarchical reinforcement learning, the top-level dynamic programming-based optimal task scheduling avoids infeasible objectives during training and simultaneously achieves dynamic multi-objective optimization for long-haul HHT operation.

The remainder of this paper is organized as follows: Section 2 describes the train model, task model and optimization model used. Section 3 details the design of the MTC-HRL method. Section 4 verifies the effectiveness of the proposed method through simulation case studies.

2 | Analytical Model

This section first develops a dynamic model of HHT. It then analyses operational strategies for both long steep downgrades and rolling terrain, establishes an event-driven multi-task classification model, and formulates differentiated optimization models for distinct operational tasks.

2.1 | Longitudinal Dynamics Model of HHT

The HHT is composed of a mixed formation with n_m locomotives and n_l wagons. This paper takes a typical 20,000-ton HHT in China as an example. Such a train typically consists of one locomotive hauling 108 unpowered freight cars, followed by another locomotive hauling an additional 108 unpowered freight cars, resulting in a total train length exceeding 2 km. The two locomotives employ wireless communication to achieve synchronized control and coordinated operation of the entire train. The train's pneumatic brake system features a single air pipe running through all locomotives and cars. Braking application and release are achieved by controlling the charging and venting of this pipe. During braking or release, the action propagates sequentially along the direction of travel via an air wave. The structure of the 20,000-tonne train is illustrated in the accompanying Figure 1.

The dynamics of the train are characterized by minor differences in the motion states (such as speed and acceleration) among individual locomotives and freight cars due to external environmental conditions and inter-vehicle interactions, while maintaining overall consistent behaviour. The longitudinal dynamics can be formulated as shown in Equation (1).

$$m_i \dot{v}_i = \alpha f_i^t - (1-\alpha) f_i^{et} - f_i^B + f_{i-1}^c - f_i^c - f_i^r,$$
$$i = 1, 2, \ldots, n = n_m + n_l \tag{1}$$

where f_i^t and f_i^{et} is the traction and electric braking force of each car. For HHT, only locomotives can output traction electric braking force, while freight cars can only output air braking force. And a switching variable is designed, where $\alpha = 1$ represents locomotive traction condition, $\alpha = 0$ represents locomotive electric braking condition.

The running resistance of HHT f_i^r in Equation (1) includes basic resistance w_i^b (given by Davis equation), gradient additional resistance w_i^g, and curve additional resistance w_i^c, which are analytically expressed in Equation (2).

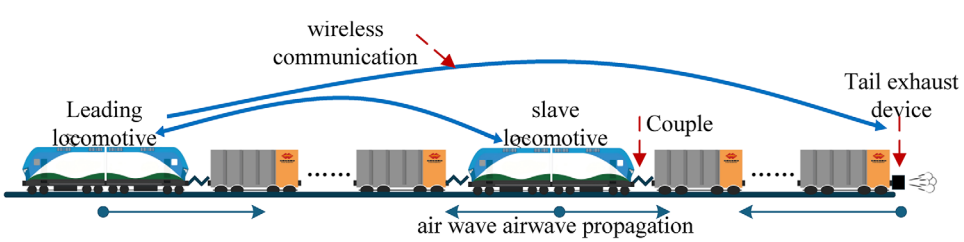


FIGURE 1 | The structure of the 20,000-ton HHT.

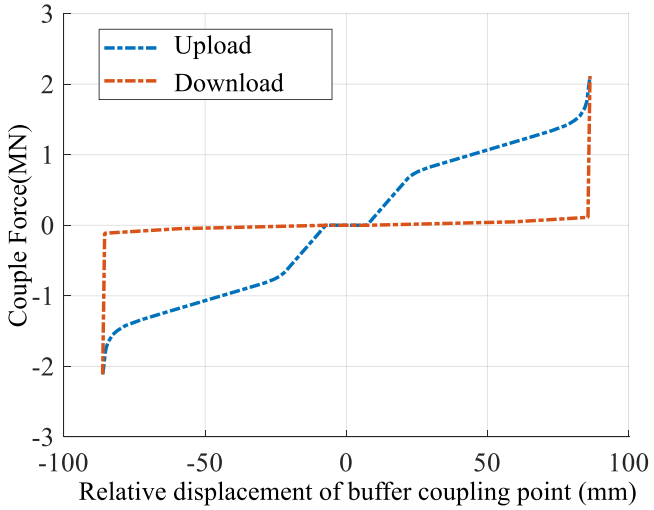


FIGURE 2 | The force–displacement (F-D) characteristics.

$$
\begin{aligned}
f_i^r &= w_i^b + w_i^g + w_i^c \\
w_i^b &= m_i \left(c_0 + c_1 v_i + c_2 v_i^2\right) \\
w_i^g &= m_i g \sin\alpha \\
w_i^c &= \frac{m_i g \cdot a}{R},\ a = 600
\end{aligned}
\tag{2}
$$

The impact forces f_i^c generated by the relative displacement between car bodies are absorbed and transmitted through these coupler and draft gear mechanisms. Accordingly, it is necessary to establish the force characteristics of the coupler and buffer system to describe the variation of the coupler force with the interaction between vehicles, that is, the force–displacement (F-D) characteristics. This paper employs a lookup table to represent the F-D characteristics, typically utilizing two fixed tables to match the buffer behaviour [47]: one fixed table for the loading process, referred to as the loading table (or loading curve), and another fixed table for the unloading process, referred to as the unloading table (or unloading curve). An intermediate slope can bridge the loading and unloading curves during buffer switching. Furthermore, the impact velocity is introduced to characterize the transition process and to smooth the switching between loading and unloading phases.

Equation (3) is used to calculate the longitudinal coupler force, as shown in Figure 2.

$$
F(x_t, v_t) =
$$

$$
\begin{cases}
f_u(x_t), & |v_t| > v_e \cap v_t x_t \geq 0, \\
f_l(x_t), & |v_t| > v_e \cap v_t x_t < 0, \\
f_l(x_t) + (F(x_{t-\Delta t}, v_{t-\Delta t}) - f_l(x_t))\frac{|v_t|}{v_e}, & |v_t| \leq v_e \cap v_t x_t \geq 0, \\
f_u(x_t) + (F(x_{t-\Delta t}, v_{t-\Delta t}) - f_u(x_t))\frac{|v_t|}{v_e}, & |v_t| \leq v_e \cap v_t x_t < 0.
\end{cases}
\tag{3}
$$

where, x_t represents the relative displacement between vehicles, v_t represents the relative speed, $f_u(x_t)$ represents the loading curve, $f_l(x_t)$ represents the unloading curve, and v_e represents the impact velocity

The air braking force f_i^B is calculated according to the traction calculation regulations and is given in Equation (4) as the product of the brake shoe pressure and the friction coefficient of each freight car.

$$
\begin{aligned}
K_i &= \frac{\pi \cdot d_z^2 \cdot p_z \cdot \eta_z \cdot n_z}{4 \cdot n_k \cdot 10^6} \\
f_i^B &= \varphi_i \cdot K_i
\end{aligned}
\tag{4}
$$

where, p_z represents the air pressure in the brake cylinder. As this paper primarily uses a 50 kPa pressure reduction as an example, p_z can be considered a fixed value. φ_i denotes the friction coefficient, while d_z, η_z, n_z and n_k are parameters related to the structure of the brake system. For specific types of locomotives and rolling stock, these values can be obtained by referring to lookup tables.

2.2 | Event-Driven Temporal Dependency Task Classification Model

This section conducts a focussed analysis on the operational regimes of HHTs. In accordance with distinct control strategies, operational tasks are systematically categorized, and a task classification model is established to support subsequent targeted optimization approaches.

Owing to their primary function in freight transportation, HHTs generally operate over extended distances with significant regional environmental variations, as illustrated in Figure 3. Using a 20,000-ton HHT as a representative case, the following observations are made:

On gentle track sections, the vertical profile is characterized by continuous ascending and descending gradients. Even though trains frequently pass through vertical curve points, the absolute gradient values are generally small, resulting in limited longitudinal forces on the train. Precise speed control can be achieved solely through adjustments in traction and electric braking, without the need for additional control forces. Therefore, the operational mode under such track conditions is classified as single traction/electric braking control. The operational objectives focus on meeting predetermined sectional travel time and average speed requirements, while maximizing the utilization of gravitational potential energy to reduce traction energy consumption.

On steep downhill sections, the significant gradient generates considerable inertial forces that exceed the control capacity of the locomotive's electric braking system alone. Therefore, air braking must be additionally applied to maintain effective speed regulation, leading to the definition of a combined electro-pneumatic braking mode under such track conditions, as illustrated in Figure 4. During the initial phase of air brake application and release, the forces on each vehicle change with a time lag due to the propagation of the air wave, which can induce substantial longitudinal impulses within the consist. Consequently, the operational objective shifts toward maintaining train stability, emphasizing the optimal selection of air braking initiation timing

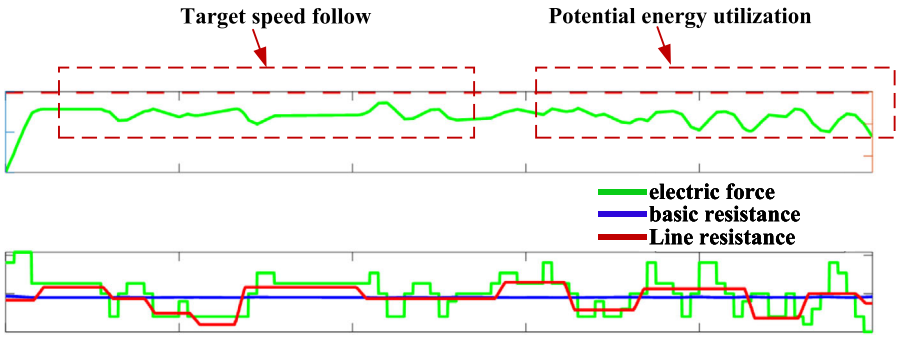


FIGURE 3 | Operating mode on gentle track sections.

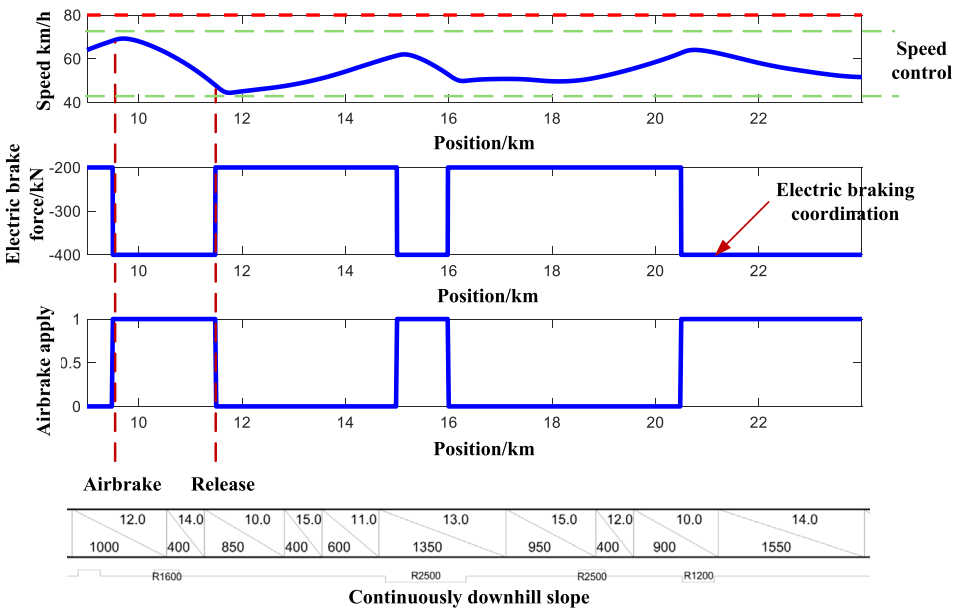


FIGURE 4 | Long steep downhill operating mode.

TABLE 1 | Task attributes in transportation systems.

Task type	Combined electro-pneumatic braking control	Single traction/Electric braking control
Line main features	Continuous slope spanning 4 km with gradient exceeding 10‰	Undulating slope with gradient below 10‰
Control mode	Air brake and electric braking	Traction/electric braking
Task target	Coupler force stability, speed safety	Speed following performance, energy efficiency
Task limit	Minimum operating speed, release time	Control force change rate

for effective speed control, while coordinating with electric braking forces to mitigate internal dynamic impacts.

Based on the analysis of train control strategies, the operational modes are categorized into two independent tasks, as summarized in Table 1.

This paper decomposes the global problem into k sub-parts, each regarded as a task segment. These tasks are categorized into two types: combined electro-pneumatic braking (abbreviated as combined braking task) and single traction/electric braking control (abbreviated as single braking task). Thereby, the global optimization problem is transformed into a series of sub-optimization problems, reducing the computational complexity inherent in solving large-scale optimization problems.

However, due to the specific operational scenarios and control conditions of each task, coupled with the fact that subsequent

tasks are influenced by the execution results of current tasks and safety constraints, a sequential dependency is presented. This necessitates the establishment of explicit event-driven task switching rules, as detailed below.

(a) Combined braking task switched to single braking task: The application time of air braking shall not be less than the specified duration $T_{b,\min}$ to ensure full development of braking performance. Meanwhile, the speed at the task endpoint must remain below the maximum permitted initial speed, which corresponds to the minimum initial speed $v^t_{0,\max}$ that prevents overspeed during the execution of the single braking task.

(b) Single braking task switched to combined braking task: The release time of air braking shall not be less than the specified duration $T_{r,\min}$ to ensure complete recharging of the brake system for subsequent applications. Meanwhile, the speed at the task endpoint must exceed the minimum permitted speed $v^b_{\min}$, which is defined to prevent excessive longitudinal impulses during brake release under low-speed conditions in combined braking task.

It should be emphasized that the task switching constraints are derived from real-world operational procedures, such as limitations on air-brake application and relief times, while the speed restrictions are imposed to ensure that the speed remains within limits throughout the journey, which aligns with actual safety requirements on site. These constraints will be quantitatively incorporated into the optimization model in Section 2.3.

2.3 | Multi-Objective Optimization Model

Based on the analysis of operational objectives and constraints for different tasks, these requirements are translated into optimization objectives and constraints for each subtask. For the single braking task, the objectives include minimizing energy consumption, target speed deviation and coupler forces. The corresponding objective function and constraints are formulated as Equation (5).

$$\min_{u_t} \quad J_t = \sum_{i=1}^{M}\left[\sum_{j=1}^{N}\left(w_v\left(v_{i,j}-v_t\right)^2 + w_e\left(\left(f_i^t-\eta_t f_i^{et}\right)\Delta x\right) + w_f\left(f_{i,j}^c\right)^2\right)\right] \tag{5}$$

subject to:

$$\begin{aligned}
&f_i^t = F^t\left(f^t_{\max}(v_i), u_i^t\right)\\
&f_i^{et} = F^{et}\left(f^{et}_{\max}(v_i), u_i^{et}\right)\\
&f_{\min} \le f_{i,j}^c \le f_{\max}\\
&v_{\min} \le v_{i,j} \le v_{\lim}\\
&v^b_{\min} \le v_{i,M} \le v_{\lim}\\
&v_{i,0} \le v^t_{0,\max}\\
&0 \le u_i^t \le 1\\
&0 \le u_i^{et} \le 1
\end{aligned} \tag{6}$$

In constraint set 6, u_i^t and u_i^{et} denotes the traction/electric braking ratio, $f^c_{i,j}$ and $v_{i,j}$ denotes the coupler force and speed of each vehicle under discrete step, v_t denotes the target speed, Δx denotes the discrete step size, M and N represent the optimization domain and the number of vehicles, respectively.

For the combined braking task, the optimization objectives are shifted toward minimizing the coupler force, target speed deviation and cumulative braking time. Furthermore, to accommodate spatial domain optimization, the minimization of cumulative braking time is transformed into the minimization of cumulative braking distance, as Equation (7).

$$\min_{u_b,u_t} \quad J_b = \sum_{i=1}^{M}\left[\sum_{j=1}^{N}\left(w_v\left(v_{i,j}-v_t\right)^2 + w_b\left(u_i^b\Delta x\right) + w_f\left(f_{i,j}^c\right)^2\right)\right] \tag{7}$$

subject to (6) and

$$\begin{aligned}
&\sum_{i=1}^{M}\left(u_i^b\Delta x\right) > S_{b,\min}\\
&v^b_{\min} \le v_{i,j} \le v_{\lim}\\
&u_i^b = [0,1]
\end{aligned} \tag{8}$$

In constraint set (8), $S_{b,\min}$ is the minimum braking distance approximated by $T_{b,\min}$.

The sub-optimization problems require the top-level optimization scheduler to select the optimal sub-problem for solution. Therefore, the top-level optimization objective is defined as the common component of both independent tasks, namely minimizing target speed deviation and average coupler force while satisfying all constraints. Furthermore, the optimization domain of the top-level problem is no longer the spatial domain but rather the segment index, defined as Equation (9).

$$\min_{\text{task}_i} \quad J = \sum_{i=1}^{K}\left[\sum_{j=1}^{N}\left(W_{V,i}\left(V_{i,j}-V_{t,i}\right)^2 + W_{F,i}\left(F_{i,j}^c\right)^2\right)\right] \tag{9}$$

subject to constraint set (6) and (8): where J represents the global objective function to be minimized, K represents the total number of task segments, N represents the total number of locomotives and wagons, $V_{t,i}$ represents the average target speed at the i-th segment, $V_{i,j}$ and $F^c_{i,j}$ represents the average speed and couple force for j-th vehicle at the i-th segment, $W_{V,i}$ and $W_{F,i}$ represents the weight for speed tracking error and coupler force.

In summary, the relationship among the three optimization models is structured as follows: the global optimization model ensures the optimal selection of tasks across sub-sections, with its objective defined in terms of the average performance metrics within each sub-section to guarantee global optimality. The sub-section optimization models are designed to achieve optimal performance under specific task requirements. This hierarchical optimization architecture enables the dynamic segmentation and optimization of the train's operation over long distances based on environmental conditions and temporal state transitions.

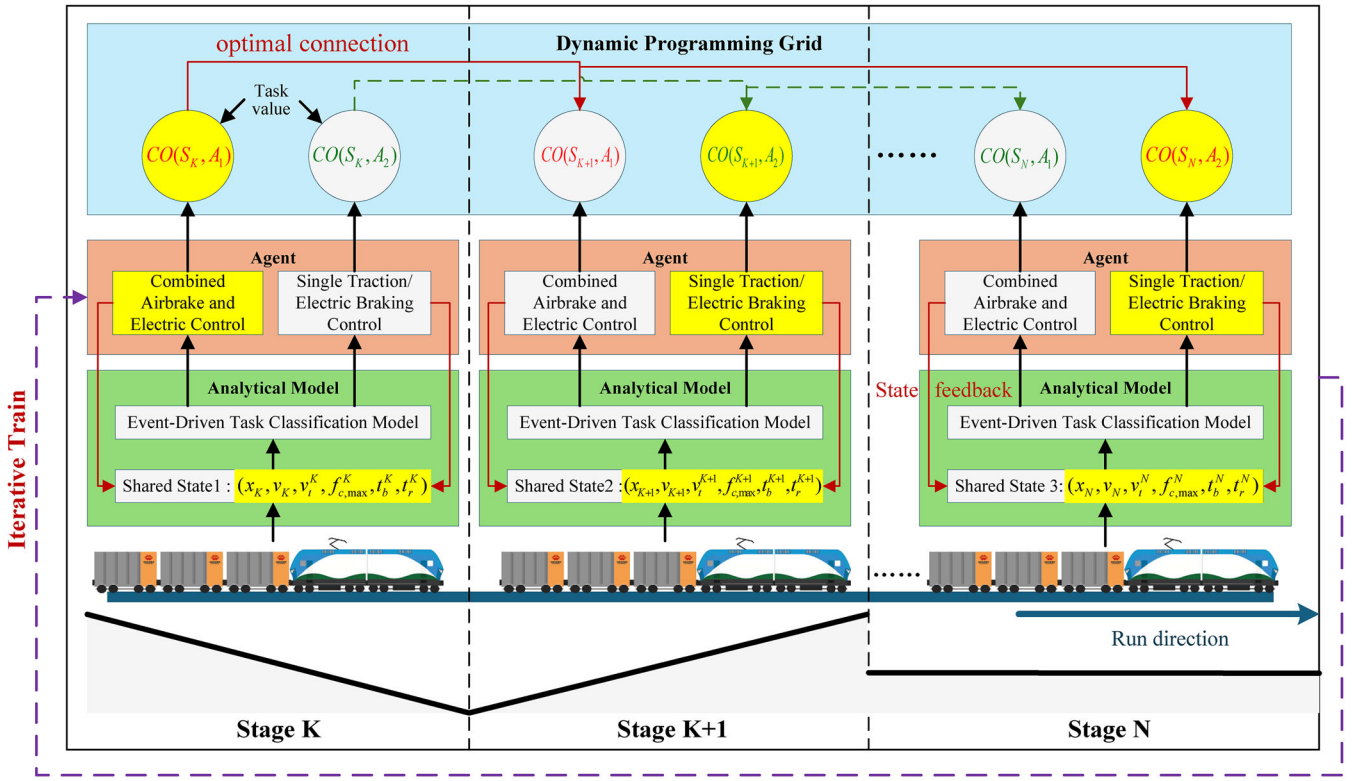


FIGURE 5 | Sequential multi-task collaboration hierarchical reinforcement learning framework.

3 | Sequential Multi-Task Collaboration Hierarchical Reinforcement Learning Method

The proposed framework follows a sequential task assignment and execution collaboration mechanism, where segment-specific subtask policies are invoked in sequence rather than being executed concurrently in a shared environment at the same decision stage.

This section first formulates the differentiated optimization problems under various subtasks as respective reinforcement learning problems. A sequential MTC-HRL framework is then designed to achieve coordinated optimization across subtasks, as illustrated in Figure 5. At the top level, the global optimization problem is transformed into a multi-stage dynamic programming problem, where optimal subtasks are selected based on stage cost values. The bottom level is responsible for training reinforcement learning policies for each subtask; here, segment-specific policies are trained independently while sharing a common state representation. By continuously training these subtask policies, the top-level cost is iteratively updated. When the bottom-level policies converge, the optimal subtask and its corresponding optimal policy are simultaneously derived, thereby achieving long-distance dynamic objective optimization.

3.1 | Reconstruction of MDP Problems for Subtasks

This subsection proceeds to formulate the reinforcement learning problem, with a primary focus on elucidating the transformation process of key reinforcement learning elements.

(a) State space: Given that sequential multi-task collaboration requires state continuity across successive subtasks, a shared state space must be designed to preserve route status information and policy execution continuity during task transitions. Consequently, the observable states of the train are constructed as a multidimensional state space, encompassing speed v, position x, cumulative braking time t_b, cumulative release time t_r, maximum coupler force $f^c_{\max}$ and target speed v_t. This state representation s_k at kth stage is formally defined as Equation (10).

$$s_k = \left[x_k, v_k, v_t^k, f^c_{\max,k}, t_b^k, t_r^k\right]. \tag{10}$$

It is important to note that the inclusion of historical braking time t_b and relief time t_r in this study is motivated by the fact that, in order to ensure the effective application of braking force and to prevent performance degradation in subsequent air brake operations, the activation and relief of air braking must satisfy minimum time constraints. By explicitly encoding cumulative braking and relief times into the state representation, the key internal dynamics of the air brake system become fully observable to the agent. This enables accurate perception of air brake status during sequential decision-making, provides state references and boundary constraints for task switching and action selection, prevents infeasible operations and accelerates convergence. By incorporating such historical variables, the temporal cumulative constraints can be transformed into a standard Markov process.

(b) Action space: To balance engineering interpretability with learning efficiency, both traction and braking commands

are discretized into uniformly spaced levels based on the maximum available traction or braking force. Each discrete action corresponds to a fixed force level, which aligns with the actual control notches of locomotives while also reducing the exploration complexity.

For single braking tasks, the action space a_k^t is designed as a bounded variable in $[-1, 1]$, representing the application ratio of electric braking to traction force as formulated in Equation (11), discretized into 21 discrete actions at an interval of 0.1.

For combined braking tasks, an integer variable in $[0, 1]$ is incorporated to indicate whether an air braking command is applied, thereby encoding it as a hybrid action a_k^b, as formulated in Equation (12).

Moreover, since only electric braking is permitted in the combined braking task, the action boundary is reduced to $(-1, 0)$. The action set is defined as a composite action space, consisting of discretized electric braking ratios at intervals of 0.1 and an integer variable indicating whether air braking is applied.

$$a_k^t = u_t, u_t \in (-1, 1) \tag{11}$$

$$a_k^b = [u_t', u_b], u_t' \in (-1, 0), u_b = [0, 1] \tag{12}$$

where, $u < 0$ indicates the application of electric braking force, while $u > 0$ represents the application of traction force.

(c) Environment: The environment employs the train longitudinal dynamics model established in Section 2.1. It should be noted that while the discrete time steps between multi-stage phases in reinforcement learning are typically set relatively large to limit the number of decisions per episode, the numerical integration of the longitudinal dynamics model requires a finer resolution. To address this, this paper adopts an implicit state formulation by inserting multiple equidistant sub-segments between adjacent decision phases as discrete computational steps for the dynamics model, as shown in Equation (13).

$$\Delta X = n\Delta x \tag{13}$$

where, n represents the sub-segments number and the train control force remains unchanged in sub-segments.

(d) Reward: To achieve multi-objective learning for the segment-specific subtask policies and enable the exploration of long-term goals, it is necessary to design a reward function that ensures smooth switching among rewards under multiple objectives. According to the optimization goals of the task, the reward function incorporates velocity tracking reward R_v, energy consumption reward R_e, air braking reward R_b and coupler force reward R_f. Specifically, R_v is defined based on the deviation between the current speed and the target speed as formulated in Equation (14); R_e represents the combined effect of traction energy consumption and regenerative energy consumption as formulated in Equation (15); R_b is decomposed into an immediate braking reward and a historical mitigation condition reward as formulated in Equation (16); and R_f is expressed as the ratio of the maximum coupler force of the entire vehicle to the upper limit of the coupler force as formulated in Equation (17).

$$R_v = \begin{cases} -\left(1 - \exp\left(-k_h\left(v - v_{\text{target}}\right)\right)\right) & v_{\text{target}} < v < v_{\text{limit}} \\ -\left(1 - \exp\left(-k_l\left(v_{\text{target}} - v\right)\right)\right) & v_{\min}^b < v < v_{\text{target}} \end{cases} \tag{14}$$

$$R_e = -\left(f_i^t - \eta_t f_i^{et}\right)\Delta x \tag{15}$$

$$R_b = \begin{cases} 0 & t_b < t_{b,\max} \\ -\left(1 - \exp\left(-k_b\left(t_b - t_{b,\max}\right)\right)\right) & t_b >= t_{b,\max} \\ 0 & t_r >= t_{r,\min} \\ -\left(1 - \exp\left(-k_r\left(t_r - t_{r,\min}\right)\right)\right) & t_r <= t_{r,\min} \cap t_b > 0 \end{cases} \tag{16}$$

$$R_f = -\left(\exp\left(-\frac{f_{k,\max}^c}{F_{\max}^c}\right)\right). \tag{17}$$

In addition to the aforementioned state-dependent rewards, a behavioural penalty C_{action} is introduced to guide the agent toward selecting more optimal actions and accelerate convergence. Within the ideal range, no penalty is imposed on actions. Beyond this range, the behavioural penalty increases with the extent to which the current action deviates from the ideal range (i.e., acceleration), as illustrated in Figure 6.

Therefore, the multi-objective comprehensive reward during the policy exploration process is represented as shown in Equation (18).

$$R^k = \alpha\left(w_v R_v^k + w_e R_e^k + w_b R_b^k + w_f R_f^k\right) - (1 - \alpha)C_{\text{action}}^k \tag{18}$$

(e) Policy: An ε-greedy exploration policy is adopted for action selection throughout the training process in both tasks. With probability ε, a random action is chosen, while with probability $1-\varepsilon$, the action a^* that maximizes the estimated policy network is selected, as defined in Equation (19).

$$\pi_\varepsilon(a|s_k) = \begin{cases} (1 - \varepsilon_0) \cdot g^{ep/ep_T} \\ \quad a = a^* = \arg\max_{a' \in \ell_a} Q(s_k, a'; \theta) \\ \varepsilon_0 \cdot g^{ep/ep_T} \\ \quad a = \text{rand}(\ell_a(s_d)) \end{cases} \tag{19}$$

where ep_T denotes the exploration rate decay frequency, g represents the decay rate of exploration rate.

3.2 | MTC-HRL Solution Method

For the sub-optimization problems within each sub-section, the training phase of this paper employs D3QN to iteratively solve the subtask reinforcement learning problems. It should be noted that D3QN is not an independent algorithm but rather an integration of two enhancement techniques: Duelling deep Q-network (duelling DQN) and double deep Q-network (DDQN), designed to address issues of Q-value overestimation and redundant Q-value estimation.

Specifically, double DQN utilizes the online network $Q(s, a; \theta)$ to select the optimal action, while the target network $Q'(s, a; \theta^-)$

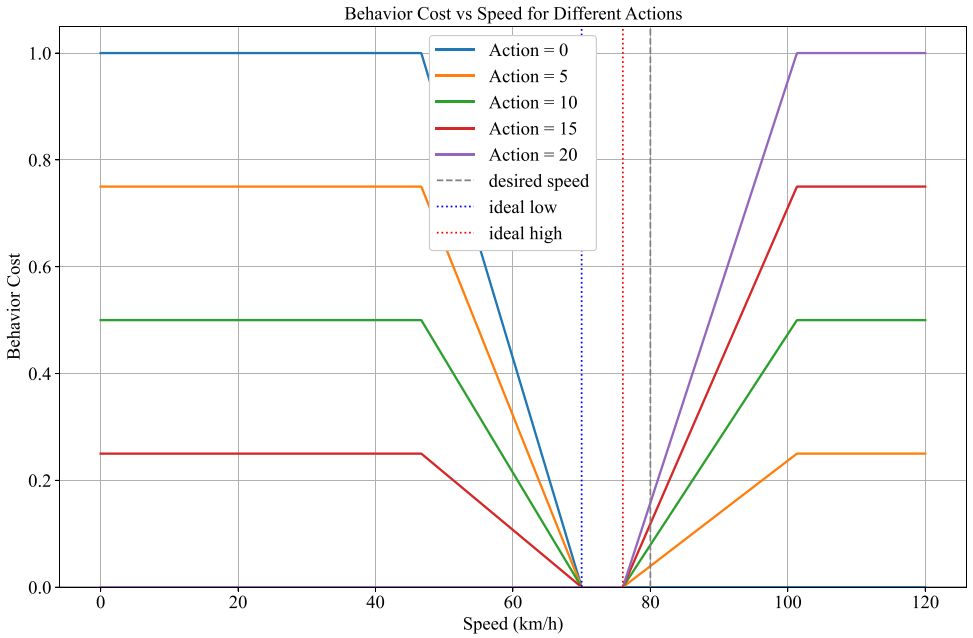


FIGURE 6 | Behavioural cost cluster.

is employed for evaluation. If the online network selects a suboptimal action due to estimation error, the independent target network is less likely to exhibit the same extreme estimation bias, thereby effectively mitigating overestimation. The iterative update rule is as formulated in Equation (20).

$$y_t^{\mathrm{DDQN}} = r_t + \gamma Q'(s_{t+1}, a; \theta^-)$$
$$a = \arg\max_a Q(s_{t+1}, a; \theta) \tag{20}$$

The network architecture of the standard DQN requires outputting a Q-value for each state-action pair, which is relatively inefficient and redundant to some extent. For instance, when a train operates on a gentle segment, multiple actions (such as traction or braking) may be similarly optimal, and their Q-value estimates show little differentiation. Significant discrepancies in the value of different actions only emerge at points of gradient change or when the environment undergoes substantial variation.

To improve efficiency, the duelling DQN is adopted by decomposing the Q-network into a value stream $V(s)$ and an advantage stream $A(s, a)$. The value stream represents the value of the state itself, that is, the average expected return of that state, while the advantage stream captures the advantage of selecting a specific action compared to the average action-effectively the deviation from the state value. This structure enables the network to learn the value of a state more effectively without requiring updates for every minor variation in action selection. When the environment is insensitive to actions, the advantage values tend to approach to zero, and the Q-value estimation is primarily governed by the value stream, leading to more stable and efficient learning, as illustrated in Figure 7.

After modifying the network architecture, the Q-value is computed as Equation (21):

$$Q(s, a) = V(s) + A(s, a)$$
$$Q(s, a; \theta, \alpha, \beta) = V(s; \theta, \beta) + \left(A(s, a; \theta, \alpha) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s, a'; \theta, \alpha)\right) \tag{21}$$

The D3QN algorithm employed in this study integrates both duelling DQN and double DQN. It is important to note that duelling DQN only modifies the network architecture specifically, the value stream and advantage stream are internal structures of the network that collectively output the Q-value. The training process does not directly update the value (V) and advantage (A) components; rather, these are learned automatically through forward and backward propagation of the network. Therefore, the training procedure of D3QN remains consistent with that of double DQN, as illustrated in Figure 8.

After the underlying sub-section optimization problems converge, the top-level framework reconstructs the global optimization model using dynamic programming to make global optimization decisions. Leveraging its 'backward update-forward selection' architecture, the framework achieves optimal subtask-policy scheduling across all sections. The key elements of the dynamic programming are defined as follows:

The state of a dynamic programming grid node is defined as a binary tuple 'section index-task type code of the previous section', $S = [K, A]$. This indicates arriving at the current decision section after executing the previous task. The task type in the current stage serves as a discrete action, equivalent to a 0-1 variable. The resulting dynamic programming grid structure is illustrated in Figure 9.

In the global optimization model, the objective of each optimization point is defined as the cost function of grid state transitions $C_O(S)$. This cost is composed of the average target speed deviation and the average maximum coupler force obtained

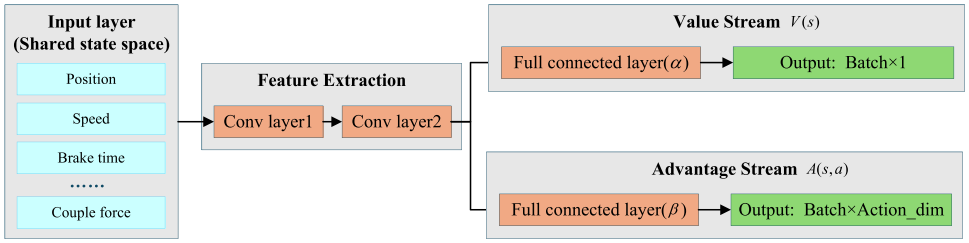


FIGURE 7 | D3QN network architecture.

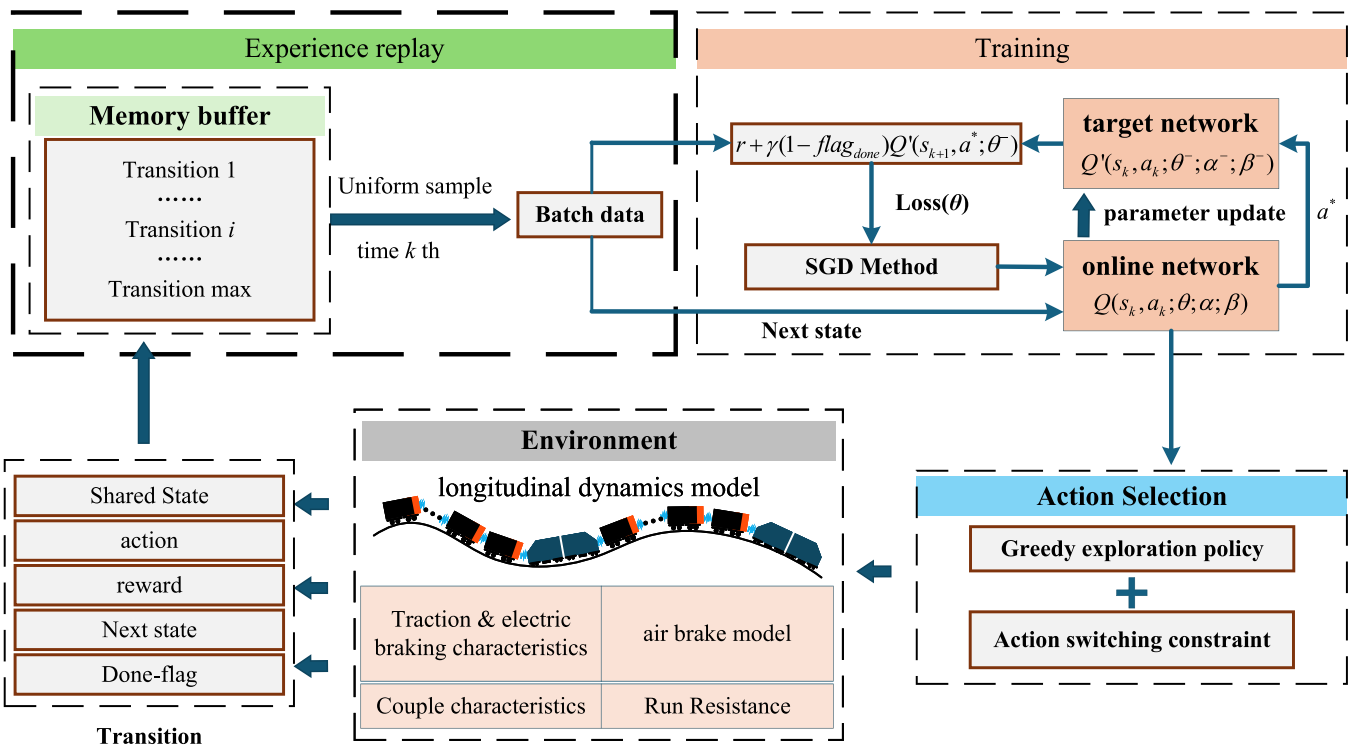


FIGURE 8 | D3QN reinforcement learning framework.

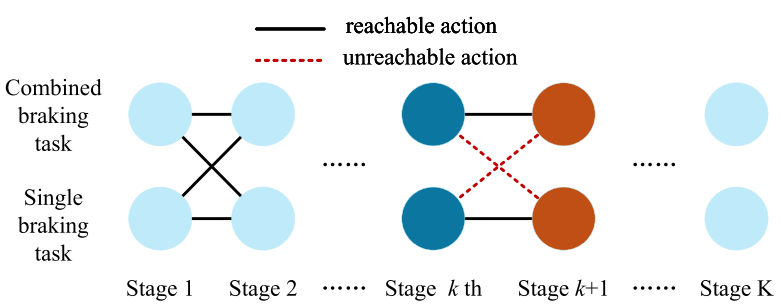


FIGURE 9 | Dynamic programming grid structure.

after executing the specified task. It should be noted that task-switching constraints are also incorporated into the cost function. If a certain task type cannot be executed under the current state, the corresponding action is assigned an infinite cost. The cost computation method is defined as follows, as shown in Equation (22).

$$C_O(S) = (V(S) - V_t(S))^2 + \Lambda(F_c(S))^2 \tag{22}$$

In summary, this subsection has presented the design principles of the underlying D3QN solving algorithm and the top-level dynamic programming framework. The two levels are interconnected through the cost function, whose specific values are derived from the outcomes of the corresponding subtask policies executing their respective current optimal decisions. The next section will provide a detailed explanation of the complete algorithmic workflows.

ALGORITHM 1 | D3QN training process for sub-task

Initialize D3QN:

D3QN online network $Q(s,a;\theta,\alpha,\beta)$ with random weights θ,α,β (Simplify to $Q(s,a;\theta)$)

D3QN target network $Q'(s,a;\theta^-,\alpha^-,\beta^-)$ with $\theta^- = \theta$, $\alpha^- = \alpha$, $\beta^- = \beta$

Experience replay buffer B with capacity N

Exploration rate $\varepsilon = \varepsilon_0$

Target network update frequency C

Initialize environment:

Route conditions (task section), vehicle parameters, task type

Train:

for episode = 1 to M **do**

Initialize random state s_0

for time step $k = 0$ to $T-1$ **do**

With probability ε select random action a_k, otherwise $a_k = \arg\max_a Q(s_k,a;\theta)$

Execute a_k, observe reward r_k and next state s_{k+1}

Store experience $(s_k,a_k,r_k,s_{k+1},\text{done})$ in B

Sample random minibatch of experiences from B

for each experience (s,a,r,s',d) in minibatch **do**

$a^* = \arg\max_{a'} Q(s',a';\theta)$

$y = r + \gamma(1-d)Q'(s',a^*;\theta^-)$

Update θ,α,β to minimize $(y - Q(s,a;\theta))^2$

end for

Every C steps: $\theta^- \leftarrow \theta$

Decay ε

$s_k \leftarrow s_{k+1}$

end for

end for

Output:

Average speed V and average maximum coupler force F_C.

Update the cost function $C_O(S,i)$ of the corresponding task i under the top-level task section S.

3.3 | Algorithm Flow of MTC-HRL

This subsection presents the detailed implementation workflow of the underlying subtask D3QN learning process in MTC-HRL and the top-level dynamic programming, primarily illustrated in pseudocode form.

First, the underlying D3QN primarily addresses the learning of optimal policies for different tasks under random initial states and stochastic environmental conditions. The detailed pseudocode for its training phase is presented in Algorithm 1.

After all underlying subtask policies have converged across different sections and various initial states, the corresponding policy can be invoked whenever its designated task is selected. In other words, once the cost functions between all grid states in the top-level framework are fixed, the dynamic programming algorithm is called to perform an optimal task sequence search. The algorithmic procedure is outlined as Algorithm 2.

ALGORITHM 2 | Hierarchical optimization.

Initialize:

Dynamic programming grid cost $C_O(S_i) = \inf,\ i = 1,2,\ldots,K$

task type task_k

Optimal search:

for each task segment $i = 1:K$ **do**

call policy_b and policy_t using D3QN algorithm

Observe performance metrics $V(S_i,\text{task}_k)$ and $F_c(S_i,\text{task}_k)$

if task_k at S_i is infeasible according to classification model **then**

$$C_O(S_i,\text{task}_k) = \inf.$$

else

else $C_O(S_i,\text{task}_k) = (V(S_i,\text{task}_k) - V_t(S_i,\text{task}_k))^2 + \Lambda(F_c(S_i,\text{task}_k))^2.$

end if

end for

Output:

for task segment $i = 1:K$ **do**

select $\text{task}_k = arg\max_{\text{task}_k} C_O(S_i,\text{task}_k)$

Dynamic programming output:

Record optimal state S_i and action A_k

end for

4 | Experimental Results and Analysis

Based on the analytical model established in Section 2 and the hierarchical reinforcement learning algorithm proposed in Section 3, this section designs simulation experiments to validate their effectiveness. The experimental scenario is built upon actual parameters from a domestic heavy-haul railway line, aiming to verify the performance of the proposed method when applied to the control of a 20,000-ton HHT. The validation consists of the following aspects: First, within a single task segment, the effectiveness of the integrated pneumatic and electrical braking control strategy versus traction-only or electrical braking-only control strategies under different initial states is verified. This ensures that each task segment possesses state-independent endpoint properties, thereby enabling flexible concatenation of tasks. Second, the training convergence of the algorithm under varying task segment lengths is analysed to examine the contribution of the multi-subsegment decomposition optimization method to improving long-distance optimization efficiency. Finally, an analysis of long-distance optimization effectiveness is conducted in a sequential multi-task collaboration setting, with comparative verification against existing planning methods and a monolithic reinforcement learning approach.

TABLE 2 | Train parameters used in the simulation.

Parameter type	Value	Parameter type	Value
Locomotive type	HXD1	Cars type	C80
Locomotive length	35.2 m	Cars length	13.2 m
Locomotive weight	200 t	Cars weight	100 t
Maximum traction force	760 kN	Maximum cars air braking force	10 kN
Maximum electric braking force	460 kN	Maximum coupler force	± 2500 kN

TABLE 3 | Core hyperparameter settings for the reinforcement learning algorithm.

Hyperparameter	Value/Description
Discount factor	$\gamma = 0.999$
Learning rate (Adam optimizer)	Dynamic learning rate scheduling strategy: initial learning rate 3×10^{-4}, minimum learning rate 3×10^{-6}, patience 50, decay factor 0.5
Experience replay buffer size	204,800
Batch size	1024
Target network update	Hard update every 20 steps
Exploration strategy	Cosine decay ε-greedy scheme: ε decays from 1.0 to 0.05
Loss function	Mean squared error (MSE) loss

4.1 | Simulation Conditions

The simulation section is based on a 0–340 km segment of a real railway line, which includes two continuous long downgrades and a section of undulating terrain. The train adopts a '1+1' configuration, where one locomotive hauls 108 freight cars, which are connected to another set of 108 freight cars hauled by a second locomotive. The specific parameters are provided in Table 2.

The core hyperparameters for the reinforcement learning algorithm are summarized in Table 3.

4.2 | Single Traction/Electric Braking Control Task

Simulations were conducted for the same sub-section (20– 40 km) under different initial speeds. With a speed limit of 80 km/h and without pneumatic braking, the resulting speed profiles and the corresponding control forces are shown in the Figure 10.

Under random initial conditions, the learned task policy consistently converges to the pre-trained optimal policy curve. In the initial phase, the policy tends to apply a larger traction force at lower speeds to achieve rapid acceleration towards the target speed, with a longer duration of traction observed at lower initial speeds. Subsequently, the policy transitions to maintaining the target speed while fully utilizing the track's potential energy, with control inputs of smaller magnitudes. Eventually, the train response tracks and stabilizes along the optimal policy curve, ensuring consistent and stable operation. These observations validate that the proposed multi-objective policy effectively balances safety, energy efficiency and stability across varying initial states.

In terms of reward outcomes, lower initial speeds correspond to longer recovery times for returning to the optimal policy, lower average speeds and consequently, lower cumulative rewards.

4.3 | Combined Air and Electric Braking Task

Simulations were conducted for a long downhill section (0–20 km) under different initial speeds, with the speed limit maintained at 85 km/h. The timing of air braking and the strategy for coordinating with electric braking force were optimized. The resulting speed profiles and the corresponding electric and air braking force performances are shown in Figure 11.

On long downhill sections, the learned task policy is highly sensitive to the initial state, with a notable divergence between the optimal strategies at low and high initial speeds. When the initial speed is low, the optimal policy employs sequential applications of air braking to control speed while ensuring compliance with brake release time constraints. In this scenario, the electric braking force must be adjusted frequently to complement the performance of the air braking system. Conversely, when the initial speed is high, the optimal policy shifts towards maintaining a prolonged braking state. Here, the electric braking force remains minimal in magnitude, thereby reducing the rate of speed reduction and achieving an extended braking distance.

4.4 | Analysis of MTC-HRL

After validating the effectiveness of the task-specific optimal policy for specific tasks, this section focuses on analysing the advantages of the hierarchical reinforcement learning

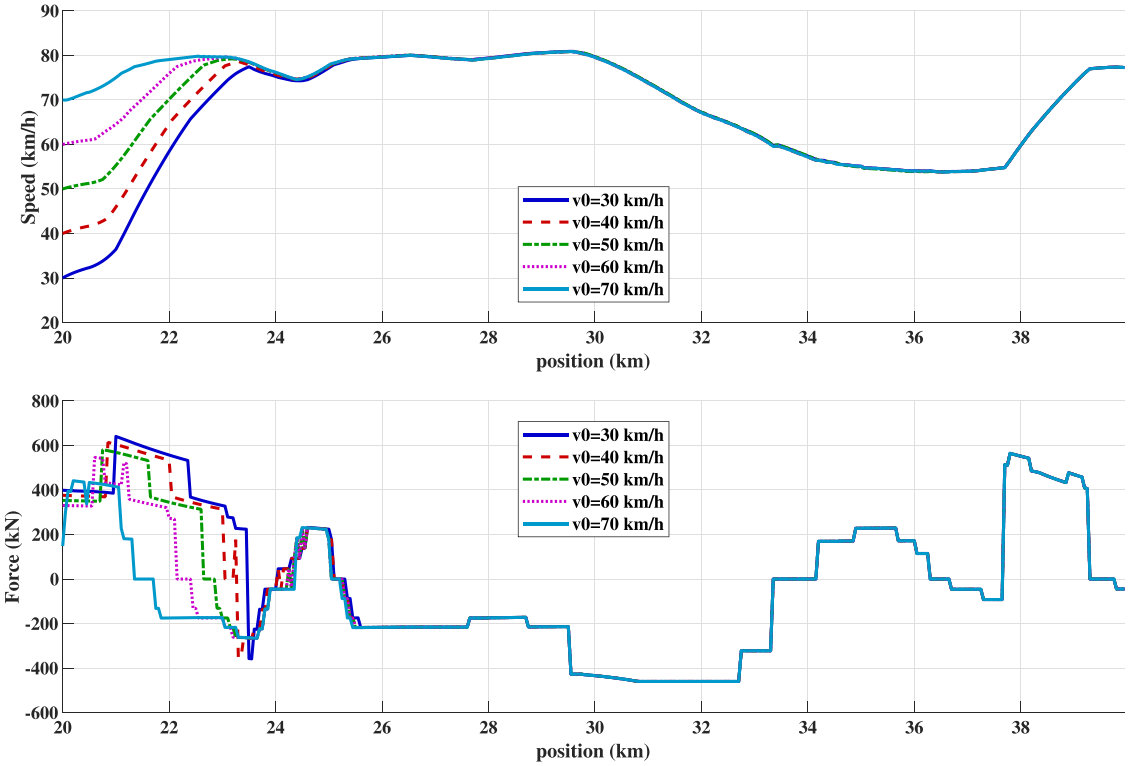


FIGURE 10 | Single braking control effect.

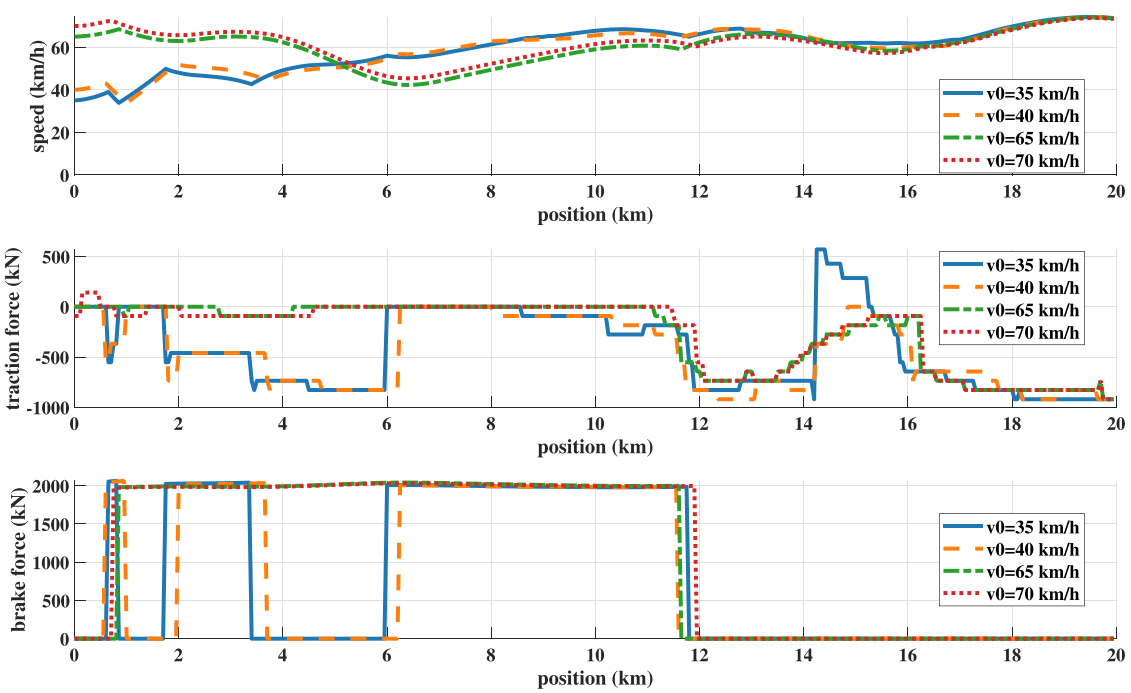


FIGURE 11 | Combined braking effect.

framework. First, the influence of task segmentation granularity is investigated by setting individual task segment lengths of 10, 20, 50 and 100 km. Taking the traction/electrical braking speed control policy as an example, the training results are presented in Figure 12.

As observed, finer task segmentation leads to a more rapid decrease in training loss and faster policy convergence. At segmentation granularities of 10 and 20 km, the policy converges after 2500 and 3200 episodes, respectively, whereas no convergence is achieved even after 5000 episodes at 50

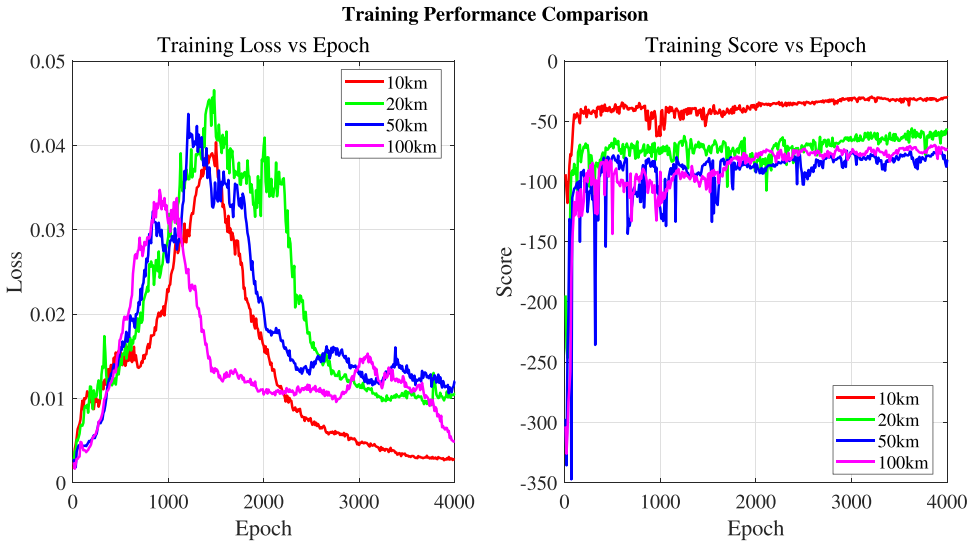


FIGURE 12 | Training effect of task policies under different task lengths.

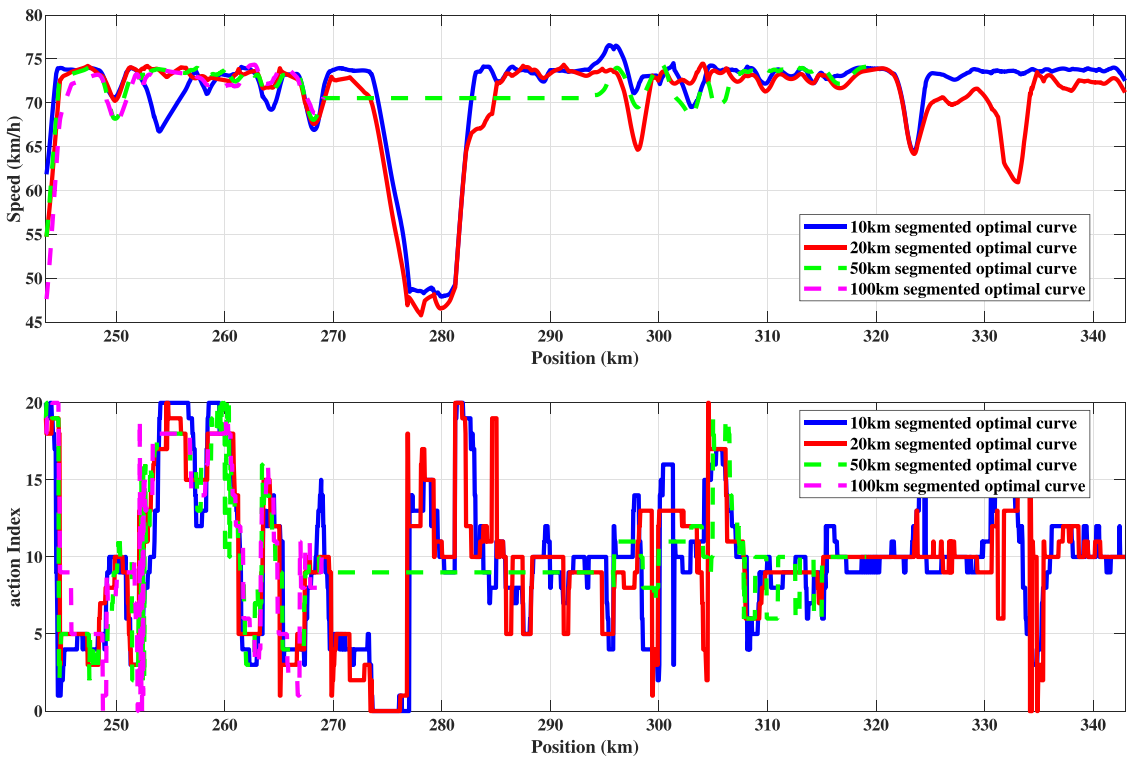


FIGURE 13 | Optimized result of task policies under different task lengths.

and 100 km granularities. In terms of rewards, higher segmentation granularity yields superior average rewards. For the actual line segment between 240 and 340 km, the traction/electrical braking speed control policy was exclusively employed for training without considering task switching, and the globally optimized curve obtained is presented in Figure 13.

Following 5000 training iterations, it becomes evident that the task segment length with a precision of 100 km is equivalent to the scope of global optimization. Similar to the case with 50 km precision, convergence has not been achieved, and an early stopping phenomenon has occurred, preventing the attainment of an effective global optimization curve.

In contrast, the trends of the global optimization curves under 10 and 20 km precisions are relatively similar, primarily because the optimization objectives and their weight settings remain unchanged. Meanwhile, given the significant emphasis on speed following in the objectives, the optimization results at a 10 km precision demonstrate a closer alignment with target speed tracking, indicating superior optimization quality. This finding

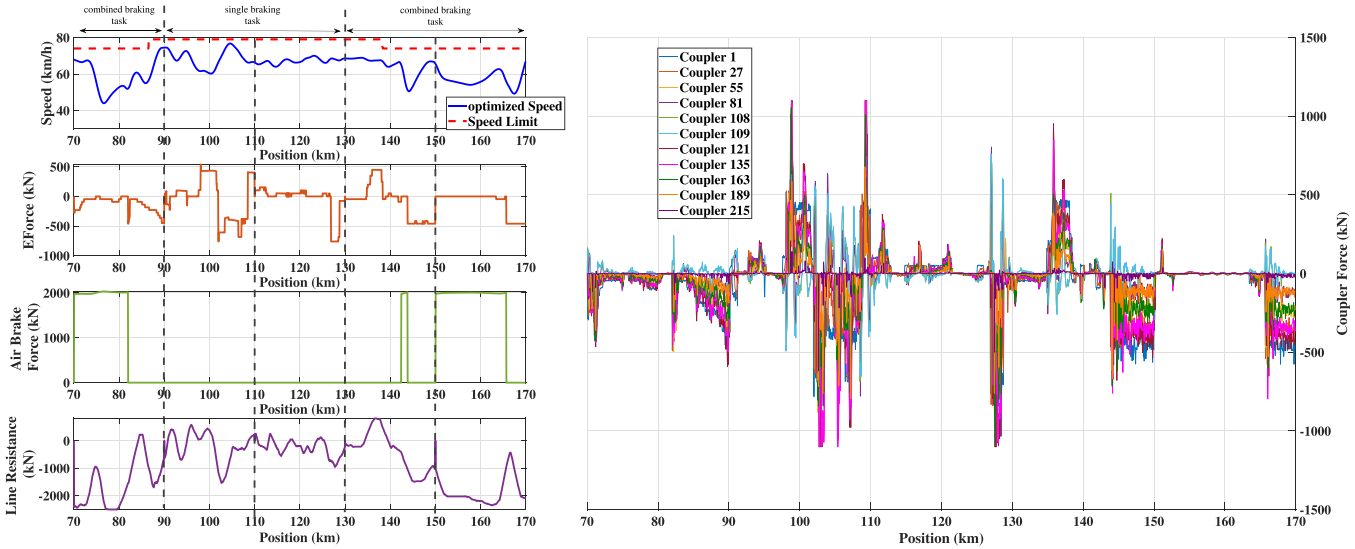


FIGURE 14 | Global optimization strategy under multi-task allocation.

aligns with the conclusion in Figure 12 that 'higher precision yields higher rewards.'

In conclusion, this study confirms that the multi-sub-segment optimization strategy outperforms the global optimization method in terms of both convergence speed and result quality.

The simulation section spans from 70 to 170 km, incorporating three consecutive long and steep downhill segments and two sections of undulating terrain. This setup is designed to validate the global optimization effect of multi-task optimal allocation over long distances. The optimization curves are illustrated in Figure 14.

The left figure depicts the optimization results for train speed and force. The line resistance increases significantly in long and steep downhill sections. In response, the optimization strategy shifts to a combined braking task, primarily utilizing air braking supplemented by electric braking, as demonstrated in the second and third subplots. As the line resistance decreases, the optimization strategy disengages air braking and transitions to a single electric braking speed control task. During this phase, the average speed approaches the target speed, while the electric braking force coordinates with the line resistance to fully harness the potential energy of the route. In the final long and steep downhill segment, the strategy reverts to combined braking.

During the integrated braking phase, the electrical braking force primarily serves a compensatory role in the overall train operation. In non-integrated braking phases, the electrical braking force is utilized to counteract track resistance to maintain constant speed, while also enabling energy-efficient coasting by leveraging track resistance.

Over long distances, and subject to meeting switching conditions, the MTC-HRL framework primarily switches tasks based on changes in line resistance, achieving dynamic multi-objective optimization in response to environmental variations. In terms of performance metrics, during the combined braking task segments, the average speed is 54.3 km/h with an average energy consumption of −459.49 kWh. In the single electric braking segments, the average speed is 63.8 km/h, accompanied by an average energy consumption of −108.85 kWh.

The figure on the right presents the optimization results of the train coupler forces, illustrating the variation in forces for selected vehicles at typical positions. During the air braking phase, the couplers are fully compressed, resulting in a clearly stratified distribution of coupler forces along the train consist. In non-air braking phases, the coupler forces are primarily influenced by the leading and trailing locomotives. When the strategy prioritizes energy efficiency, both the control and coupler forces remain relatively small. Conversely, when the strategy shifts to tracking the target speed, the control force undergoes significant short-term variations, leading to impactful shocks on the couplers. These observed phenomena align well with practical operational scenarios. Regarding performance metrics, the maximum tensile and compressive coupler forces are both constrained within 1100 kN, with average values of 72.82 and −105.91 kN, respectively.

4.5 | Performance Advantages of MTC-HRL Compared With Baseline Method

To verify the performance advantages of the proposed MTC-HRL framework, we conducted supplementary simulation comparison experiments with the baseline single-agent reinforcement learning method (D3QN). The results clearly demonstrate the proposed framework's superiority in terms of convergence speed, energy consumption and operational stability.

The optimization curves compared with the baseline method are shown in Figure 15. The traditional D3QN method must satisfy the complex physical constraints of combined air and electric braking throughout the entire operation, and its optimization objectives and weights remain fixed, making it difficult to balance the multi-objective requirements across different operational phases. In contrast, MTC-HRL adaptively adjusts the

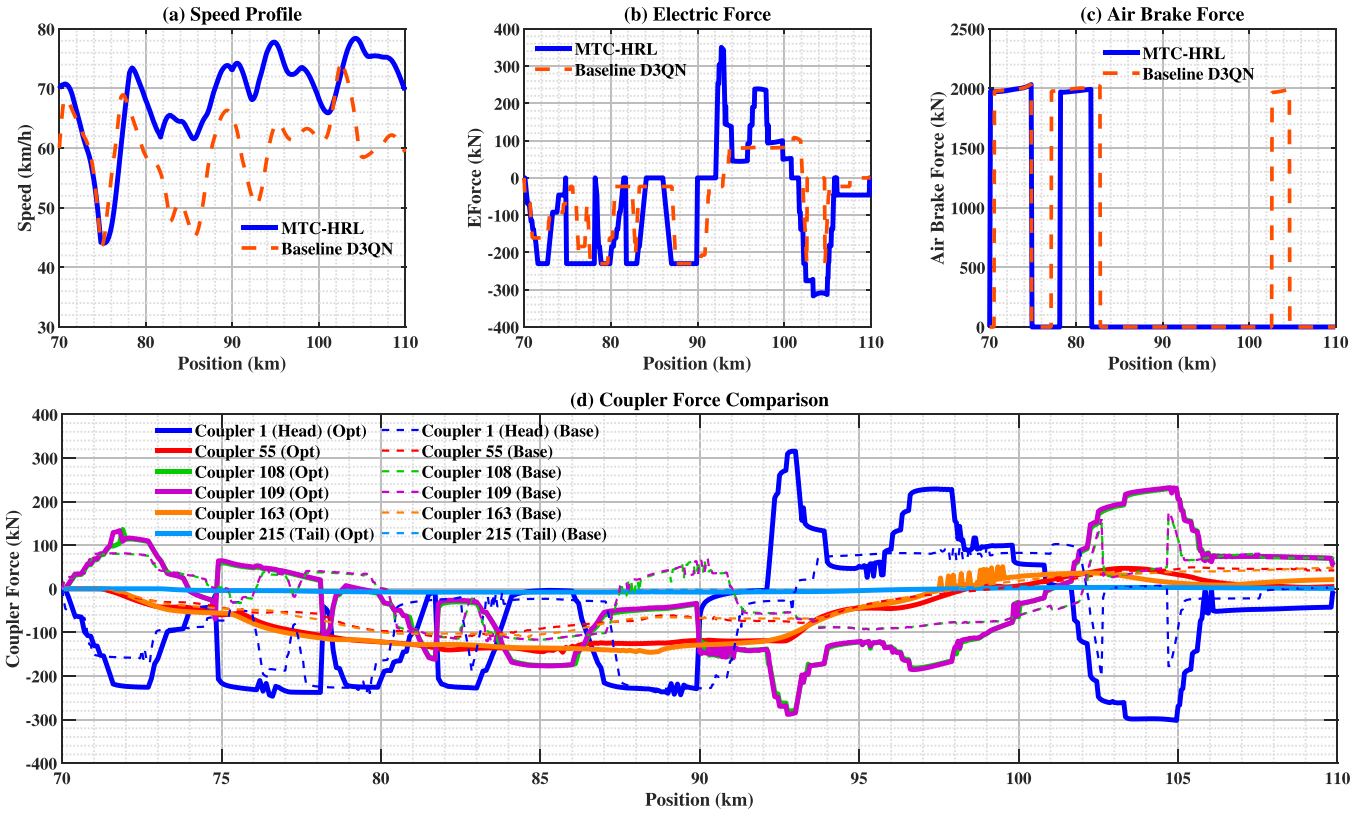


FIGURE 15 | Comparison between MTC-HRL and D3QN.

TABLE 4 | Performance comparison between MTC-HRL and baseline method.

Performance indicator	MTC-HRL	Baseline (D3QN)
Average positive coupler force (tension)	40.43 kN	34.58 kN
Maximum positive coupler force (tension)	315.67 kN	177.77 kN
Average negative coupler force (compression)	−90.32 kN	−66.92 kN
Maximum negative coupler force (compression)	−301.96 kN	−240.16 kN
Energy consumption	−1196.66 kWh	−782.38 kWh
Average speed	70.03 km/h	59.73 km/h
Air braking distance	8.60 km	12.15 km

optimization focus for different segments through multi-task coordination and a dynamic multi-objective trade-off mechanism, thereby achieving comprehensive global optimization.

Specifically, in non-air-braking segments, MTC-HRL increases the weights for punctuality and energy consumption optimization, enabling the train to maintain a higher average speed and operational efficiency. In air-braking segments, the method dynamically invokes the combined air and electric braking task based on the operational state, achieving shorter air-braking distances and a smoother deceleration process. Detailed performance indicators are compared in Table 4.

It should be noted that in non-air-braking segments, due to the higher priority given to speed and punctuality objectives, MTC-HRL employs larger control outputs with more frequent switching compared to the baseline method. Consequently, the coupler force in these segments increases, which can be interpreted as trading off a portion of ride comfort for significant speed improvement. However, the overall coupler force remains constrained within 1000 kN, satisfying the design requirements.

The convergence comparison is illustrated in Figure 16. The proposed MTC-HRL framework adopts a sequential multi-task collaboration strategy, in which different task segments are assigned dedicated subtask policies that are trained independently and coordinated through shared state representations. Since each subtask policy only needs to learn and make decisions within its corresponding task segment, the state-space dimensionality of each individual decision-making problem is significantly reduced compared to the globally unified optimization in D3QN.

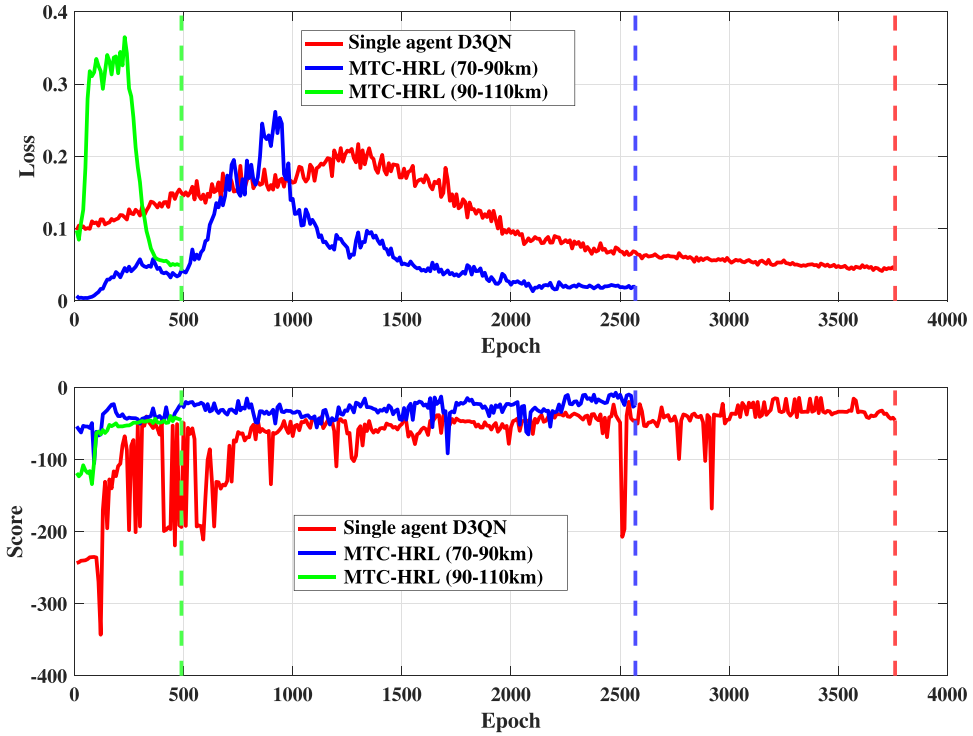


FIGURE 16 | Comparison of convergence performance.

This accelerates the training convergence speed under identical reinforcement learning hyperparameter conditions.

Furthermore, although the segment-specific subtask policies in this study are invoked sequentially during execution, they are trained independently. Thus, the overall convergence time of the algorithm can be approximated as the sum of the independent training convergence times of the policies corresponding to each task segment. Simulation results show that the baseline D3QN method typically stops automatically after approximately 3700 steps, with a convergence time of about 7200 s. Under the MTC-HRL framework, the combined braking task converges in about 2570 steps (2900 seconds), and the single electric braking task converges in about 490 steps (approximately 450 s). The total convergence time is approximately 3500 s, representing a training efficiency improvement of about 50%.

It should be clarified that the above conclusions are not based on a single experimental trial. Consistent convergence processes and time characteristics were observed across multiple simulations under identical configuration conditions. Therefore, the presented results are representative and reflect the typical training performance of the algorithm. Compared to the baseline D3QN method, the proposed framework achieves a conservative estimated improvement of at least 40% in training time.

5 | Conclusion

This paper addresses the issues of low computational efficiency and limited adaptability of existing intelligent algorithms in the long-distance multi-objective operational optimization of HHTs. A sequential MTC-HRL framework is proposed, which achieves a balance between optimization accuracy and computational efficiency through task decomposition, parallel subtask learning and sequential collaboration strategies.

The main contributions of this study are summarized as follows:

1. Event-driven task classification and hierarchical optimization framework: An event-driven, time-dependent task classification model is established, decomposing the long-distance optimization problem into a combination of electro-pneumatic combined braking control tasks and single traction/electrical braking control tasks. A hierarchical sequential collaboration architecture is designed, in which segment-specific subtask policies are trained in parallel and coordinated through top-level dynamic programming during route execution. This hierarchical decomposition strategy significantly reduces computational complexity and improves convergence performance. Simulation results show an improvement of nearly 40% in convergence episodes compared to global training.
2. Multi-objective coordinated control strategy: Differentiated objective reward functions and constraint conditions are formulated for different tasks. Based on the D3QN learning algorithm and through shared state space, smooth task transitions and sequential collaborative decision-making across subtasks are achieved. Simulation results demonstrate that coupler forces are effectively limited within 1100 kN, and both the average speed and energy consumption align with actual operational conditions. This realizes a dynamic balance among multiple objectives, including safety, stability, energy efficiency and punctuality.

Based on the conditions of a real 340-km railway line, compared to the global optimization method using single-agent reinforcement learning, the proposed MTC-HRL framework exhibits faster convergence speed and superior optimization quality, confirming its suitability for large-scale, long-distance HHT operational optimization.

Furthermore, based on the simulation results, the practical effectiveness of the proposed framework is influenced by several key hyperparameters. In particular, the segment length should be determined by balancing task complexity and training stability: overly short segments may cause frequent task switching and increased training complexity, whereas overly long segments may reduce the performance benefits of multi-task switching. Moreover, different reward weight settings may change the preference among energy efficiency, speed tracking and longitudinal smoothness, thereby affecting the corresponding multi-objective trade-offs. Benefiting from its hierarchical and segment-wise collaborative structure, the proposed framework is expected to remain effective under moderate parameter variations and to exhibit better adaptability than a single global policy under moderate changes in gradients, speed limits, and load conditions. However, more substantial environmental variations may still require retraining or online fine-tuning, which will be investigated in future work.

Future research directions include extending this framework to heterogeneous multi-locomotive coordination scenarios and designing more scenario-specific tasks to achieve efficient parallel optimization. In addition, future work may explore a control framework based on externally configurable weight parameters. This framework aims to decouple long-term system dynamics from task-specific reward functions, thereby enabling externalized configuration of optimization objective weights.

Author Contributions

Jianhua Wang: conceptualization, resources, writing – original draft. **Cong Wang:** conceptualization, methodology, visualization, writing – original draft. **Qingyuan Wang:** conceptualization, funding acquisition, supervision, writing – review and editing. **Yiding Yu:** visualization, writing – original draft, writing – review and editing. **Xiaoyun Feng:** supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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